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Vector Databases for RAG: An Introduction ✕

Today's goals

○ Complete any 3 learning items - 2/3
○ Complete a module
○ Complete a practice item

Today's Skill Points 11 XP
See skill progress

Module 1
Introduction to Vector Databases and Chroma DB

Welcome to the Course

- ✓ IBM Product Spotlight: watsonx.data
  Reading • 2 min
- ✓ Video: Course Introduction
  Video • 3 min
- ○ **Course Overview**
  Reading • 5 min
- ✓ RAG and Agentic AI Professional Certificate Overview
  Video • 5 min
- ✓ Reading: Helpful Tips for Course Completion
  Ungraded Plugin • 5 min

Introduction to Vector Databases and Similarity Search

- ○ Vector Database Concepts
  Video • 6 min
- ○ Reading: Vector Databases Versus Traditional Databases
  Ungraded Plugin • 5 min
- ○ Vector Database Types
  Video • 9 min
- ○ Applications of Vector Databases
  Video • 6 min
- ○ Reading: Similarity Search
  Ungraded Plugin • 30 min
- ○ Lab: Similarity Search by Hand
  Ungraded App Item • 30 min
- ○ [Optional] Interactive Lesson Recap Podcast (AI-Powered)
  Ungraded App Item • 10 min
- ○ Practice Quiz: Introduction to Vector Databases and Similarity Search
  Practice Assignment • 10 min

Exploring Chroma DB

- ○ Chroma DB Key Concepts and Architecture
  Video • 6 min
- ○ Reading: Chroma DB Filtering
  Ungraded Plugin • 15 min
- ○ Reading: Similarity Search and HNSW in Chroma DB
  Ungraded Plugin • 30 min
- ○ Lab: Similarity Search on Text Using Chroma DB and Python
  Ungraded App Item • 30 min
- ○ [Optional] Interactive Lesson Recap Podcast (AI-Powered)
  Ungraded App Item • 10 min
- ○ Practice Quiz: Exploring Chroma DB
  Practice Assignment • 10 min

Module Summary and Evaluation

- ○ Reading: Summary and Highlights: Introduction to Vector Databases and Chroma DB
  Reading • 3 min
- ○ Reading: Cheat Sheet: Introduction to Vector Databases and Chroma DB
  Ungraded Plugin • 15 min
- ○ Graded Quiz: Introduction to Vector Databases and Chroma DB
  Graded Assignment • 21 min
- ○ [Optional] Discussion Prompt: Meet and Greet
  Discussion Prompt • 10 min

Module 2
Vector Databases for Recommendation Systems and RAG

Course Overview

16 XP • Advanced RAG System Implementation | 8 XP • Applied Generative AI Development and Strategy

0:03 / 4:04 1x

Hello, and welcome to the **Vector Databases for RAG: An Introduction** course.

Gain expertise in using vector databases and improve your data retrieval skills in this hands-on course! Start by exploring the fundamental principles of similarity search and vector databases, understanding how they differ from traditional databases, and recognizing their importance in recommendation systems and Retrieval-Augmented Generation (RAG) applications.

Build on this foundation with practical experience using Chroma DB—one of the leading vector database solutions. Through interactive labs, learn to create collections, manage embeddings, and perform similarity searches with real-world datasets. Understand key concepts such as vector operations and database architecture to develop a strong grasp of Chroma DB's functionality.

Apply what you've learned by creating a real-world recommendation system powered by Chroma DB and an embedding model from Hugging Face. This project will enhance your understanding of how vector databases improve search and retrieval in AI-driven applications.

Throughout the course, engage in practical exercises to refine your skills in database management, similarity search techniques, and advanced vector operations. By the end, you'll have a thorough understanding of vector databases and similarity searches, which are essential components of the RAG pipeline.

This course is part of the IBM RAG and Agentic AI Professional Certificate. By completing the program, you'll understand how to differentiate between vector databases and traditional databases based on their functionality and use cases. You will be able to perform fundamental database operations in Chroma DB, including updating, deleting, and managing collections. You will develop a thorough and comprehensive understanding of key internal mechanisms within RAG. You will also learn about similarity search techniques and apply them manually and with Chroma DB, and you will be able to develop recommendation systems using these techniques.

Prerequisites

This course is accessible to a wide range of professionals, but having some knowledge and experience of the Python programming language is required. An understanding of software applications and business systems and some familiarity with databases is advantageous but not required.

Objectives

After completing this course, you will be able to:

- Differentiate between vector databases and traditional databases based on their functionality and use cases
- Execute fundamental database operations in Chroma DB, including updating, deleting, and managing collections
- Understand and apply similarity search techniques, both manually and with Chroma DB, to retrieve relevant information from real-world datasets and develop recommendation systems using these techniques
- Develop a thorough and comprehensive understanding of key internal mechanisms within RAG

Course Outline

This course consists of two modules.

Module 1: Introduction to Vector Databases and Chroma DB

- Lesson 0: Welcome
- Lesson 1: An Introduction to Vector Databases and Similarity Search
- Lesson 2: Exploring Chroma DB

Key Topics:

- Concepts of vector databases
- Comparing vector databases with traditional databases
- Types of vector databases
- Use cases and applications for vector databases
- Similarity search
- Concepts and architecture of Chroma DB
- Chroma DB filtering
- Similarity search with Chroma DB

Module 2: Vector Databases for Recommendation Systems and RAG

- Lesson 1: Chroma DB Database Operations
- Lesson 2: Develop a Recommendation System and Connect Learned Concepts to RAG

Key Topics:

- Essential Chroma DB database operations
- Analyzing data with similarity search
- How vector databases power RAG applications
- Building a recommendation system with Chroma DB

Congratulations on taking this step to boost your career with systems analysis skills. We hope you enjoy your learning journey.

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